ECE 310 Fall 2026
Lecture 8: Inverse z-transform

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Learning Objectives

- Write the common forms for the z-transform of finite-length and infinite-length signals.
- Compute the inverse z-transform of a given rational expression in the z-domain using partial fraction expansion and inspection.



Common Forms of z-transform: Finite-length Signals

Any finite-length signal $x[n]$ with start index n_s and end index n_e can be written as a linear combination of scaled and shifted Kronecker delta signals:

$$x[n] = \sum_{k=n_s}^{n_e} x[k]\delta[n-k].$$

The resulting z-transform of any discrete-time signal is then

$$X(z) = \sum_{k=n_s}^{n_e} x[k]z^{-k}$$



Common Forms of z-transform: Infinite-length Signals

Infinite-length signals have z-transforms given by *rational expressions*, i.e. a fraction with polynomials in the numerator and denominator: $X(z) = B(z)/A(z)$. When $N \leq M$, we have a *proper rational expression*.

$$X(z) = \frac{\sum_{k=0}^{N-1} b_k z^{-k}}{1 + \sum_{k=1}^{M} a_k z^{-k}} \tag{1}$$

$$X(z) = \frac{\prod_{k=1}^{N-1} (1 - q_k z^{-1})}{\prod_{k=1}^{M} (1 - p_k z^{-1})} \tag{2}$$



Inverse z-transform

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The inverse z-transform for $X(z)$ is given by

$$x[n] = \frac{1}{j2\pi} \oint_C X(z) z^{n-1} dz.$$



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$$x[n] = \frac{1}{j2\pi} \oint_C X(z) z^{n-1} dz.$$

Don't worry, we won't use this formula!



Computing Inverse z-transform: Finite-length Signals

We have already determined this rule!

$$x[n] = \sum_{k=n_s}^{n_e} x[k]\delta[n-k] \overset{\mathcal{Z}}{\longleftrightarrow} X(z) = \sum_{k=n_s}^{n_e} x[k]z^{-k}$$



Example 1: Inverse z-transform by Inspection

Determine the inverse z-transform of the following z-transforms.

(a) $X(z) = 1 - \frac{1}{4}z^{-1} + \frac{1}{16}z^{-2}$

(b) $X(z) = \frac{1}{1+\frac{1}{3}z^{-1}}, |z| < \frac{1}{3}$



Example 1: Inverse z-transform by Inspection

Determine the inverse z-transform of the following z-transforms.

(c) $X(z) = \frac{1}{1-\frac{1}{2}z^{-1}} - \frac{3z^{-1}}{1+\frac{1}{5}z^{-1}}, \ |z| > \frac{1}{2}$

(d) $X(z) = \frac{1-2z^{-1}+3z^{-2}}{(1-z^{-1})(1+\frac{2}{3}z^{-1})(1-2z^{-1})}, \ |z| > 2$



Computing Inverse z-transform: Infinite-length Signals

Consider the following third form of the z-transform for infinite-length signals:

$$X(z) = \frac{\prod_{k=1}^{N-1}(1 - q_k z^{-1})}{\prod_{k=1}^{M}(1 - p_k z^{-1})} = \sum_{k=1}^{M} \frac{A_k}{1 - p_k z^{-1}} \tag{3}$$



Computing Inverse z-transform: Partial Fraction Decomposition

We need some procedure for taking forms (1) or (2) of the z-transform in the previous slides and translating them to form (3) in the previous slide. This procedure is known as *partial fraction decomposition*.



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Partial Fraction Decomposition

Suppose we have z-transform $X(z)$:

1. Factorize the denominator of $X(z)$ to obtain "form (2)" of the z-transform for infinite-length signals and identify each pole of the system. Set this equal to "form (3)" where each pole has it's own rational expression and A_k coefficient.
2. Multiply both sides – $X(z)$ and your new factorized form – by the shared denominator of $X(z)$.
3. Solve for each A_k value by setting $z = p_k$.



Example 2

Determine $x[n]$ corresponding to the following z-transform. You may assume $x[n]$ is right-sided.

$$X(z) = \frac{2 - z^{-1}}{1 + \frac{2}{3}z^{-1} - \frac{1}{3}z^{-2}}$$



Example 3

Determine all possible ROCs for the below $X(z)$ and find the corresponding $x[n]$ in each case.

$$X(z) = \frac{1}{\left(1 - \frac{4}{3}z^{-1}\right)\left(1 - \frac{2}{3}z^{-1}\right)}$$



Example 3 (continued)

Determine all possible ROCs for the below $X(z)$ and find the corresponding $x[n]$ in each case.

$$X(z) = \frac{1}{\left(1 - \frac{4}{3}z^{-1}\right)\left(1 - \frac{2}{3}z^{-1}\right)}$$

